

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

A rectangle has a length that is **15** times its width. The function $y = (15w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $15w$ in this context?

Answer

- A. The length of the rectangle, in feet
- B. The area of the rectangle, in square feet
- C. The difference between the length and the width of the rectangle, in feet
- D. The width of the rectangle, in feet

Correct Answer: A

Rationale

Choice A is correct. It's given that a rectangle has a length that is **15** times its width. It's also given that the function $y = (15w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. The area of a rectangle can be calculated by multiplying the rectangle's length by its width. Since the rectangle has a length that is **15** times its width, it follows that w represents the width of the rectangle, in feet, and $15w$ represents the length of the rectangle, in feet. Therefore, the best interpretation of $15w$ in this context is that it's the length of the rectangle, in feet.

Choice B is incorrect. This is the best interpretation of y , not $15w$, in the given function.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of w , not $15w$, in the given function.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$f(x) = 27(1.20)^{\frac{x}{3}}$$

For the given function f , the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p ?

Correct Answer: 44

Rationale

The correct answer is 44. For a function defined by an equation of the form $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{k}}$, where a , r , and k are positive constants, the value of $f(x)$ increases by $r\%$ for every increase of x by k . The given equation can be rewritten as $f(x) = 27(1.20)^{\frac{2x}{6}}$, or $f(x) = 27\left((1.2)^2\right)^{\frac{x}{6}}$, which is equivalent to $f(x) = 27(1.44)^{\frac{x}{6}}$, or $f(x) = 27\left(1 + \frac{44}{100}\right)^{\frac{x}{6}}$. This equation is of the form $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{k}}$, where $a = 27$, $r = 44$, and $k = 6$. It follows that the value of $f(x)$ increases by 44% for every increase of x by 6. Therefore, the value of p is 44.

Alternate approach: It's given that $f(x) = 27(1.20)^{\frac{x}{3}}$. An increase in the value of x by 6 can be represented by the expression $x + 6$. The value of $f(x + 6)$ can be found by substituting $x + 6$ for x in the equation $f(x) = 27(1.20)^{\frac{x}{3}}$, which yields $f(x + 6) = 27(1.20)^{\frac{x+6}{3}}$. By the product property of exponents, this equation is equivalent to $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.20)^{\frac{6}{3}}$, or $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.20)^2$, which can be rewritten as $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.44)$. Since $f(x) = 27(1.20)^{\frac{x}{3}}$, it follows that $f(x + 6) = f(x) \cdot (1.44)$. Therefore, for any value of x , the value of $f(x + 6)$ is 1.44 times, or 44% greater than, the value of $f(x)$. It's given that the value of $f(x)$ increases by $p\%$ for every increase in the value of x by 6. It follows that the value of p is 44.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$v = -\frac{w}{150x}$$

The given equation relates the distinct positive numbers v , w , and x . Which equation correctly expresses w in terms of v and x ?

Answer

- A. $w = -150vx$
- B. $w = -\frac{150v}{x}$
- C. $w = -\frac{x}{150v}$
- D. $w = v + 150x$

Correct Answer: A

Rationale

Choice A is correct. It’s given that x is positive. Therefore, multiplying each side of the given equation by $-150x$ yields $-150xv = w$, which is equivalent to $w = -150vx$. Thus, the equation $w = -150vx$ correctly expresses w in terms of v and x .

Choice B is incorrect. This equation is equivalent to $v = -\frac{wx}{150}$.

Choice C is incorrect. This equation is equivalent to $v = -\frac{x}{150w}$.

Choice D is incorrect. This equation is equivalent to $v = w - 150x$.

Question ID: f5c3e3b8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $(m^4q^4z^{-1})(mq^5z^3)$, where m , q , and z are positive?

Answer

- A. $m^4q^{20}z^{-3}$
- B. $m^5q^9z^2$
- C. $m^6q^8z^{-1}$
- D. $m^{20}q^{12}z^{-2}$

Correct Answer: B

Rationale

Choice B is correct. Applying the commutative property of multiplication, the expression $(m^4q^4z^{-1})(mq^5z^3)$ can be rewritten as $(m^4m)(q^4q^5)(z^{-1}z^3)$. For positive values of x , $(x^a)(x^b) = x^{a+b}$. Therefore, the expression $(m^4m)(q^4q^5)(z^{-1}z^3)$ can be rewritten as $(m^{4+1})(q^{4+5})(z^{-1+3})$, or $m^5q^9z^2$.

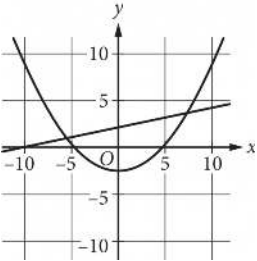
Choice A is incorrect and may result from multiplying, not adding, the exponents.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question



A system of equations consists of a quadratic equation and a linear equation. The equations in this system are graphed in the xy -plane above. How many solutions does this system have?

Answer

- A. 0
- B. 1
- C. 2
- D. 3

Correct Answer: C

Rationale

Choice C is correct. The solutions to a system of two equations correspond to points where the graphs of the equations intersect. The given graphs intersect at 2 points; therefore, the system has 2 solutions.

Choice A is incorrect because the graphs intersect. Choice B is incorrect because the graphs intersect more than once. Choice D is incorrect. It's not possible for the graph of a quadratic equation and the graph of a linear equation to intersect more than twice.

Question ID: 3c95093c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question
 $6x - 9y > 12$

Which of the following inequalities is equivalent to the inequality above?

- Answer
- A. $x - y > 2$
- B. $2x - 3y > 4$
- C. $3x - 2y > 4$
- D. $3y - 2x > 2$

Correct Answer: B

Rationale
Choice B is correct. Both sides of the given inequality can be divided by 3 to yield $2x - 3y > 4$.

Choices A, C, and D are incorrect because they are not equivalent to (do not have the same solution set as) the given inequality. For example, the ordered pair $(0, -1.5)$ is a solution to the given inequality, but it is not a solution to any of the inequalities in choices A, C, or D.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$y = x + 1$
 $y = x^2 + x$

If (x,y) is a solution to the system of equations above, which of the following could be the value of x ?

Answer

- A. -1
- B. 0
- C. 2
- D. 3

Correct Answer: A

Rationale

Choice A is correct. It is given that $y = x + 1$ and $y = x^2 + x$. Setting the values for y equal to each other yields $x + 1 = x^2 + x$. Subtracting x from each side of this equation yields $x^2 = 1$. Therefore, x can equal 1 or -1 . Of these, only -1 is given as a choice.

Choice B is incorrect. If $x = 0$, then $x + 1 = 1$, but $x^2 + x = 0^2 + 0 = 0 \neq 1$. Choice C is incorrect. If $x = 2$, then $x + 1 = 3$, but $x^2 + x = 2^2 + 2 = 6 \neq 3$.

Choice D is incorrect. If $x = 3$, then $x + 1 = 4$, but $x^2 + x = 3^2 + 3 = 12 \neq 4$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $16x^3y^2 + 14xy$?

Answer

- A. $2xy(8xy + 7)$
- B. $2xy(8x^2y + 7)$
- C. $14xy(2x^2y + 1)$
- D. $14xy(8x^2y + 1)$

Correct Answer: B

Rationale

Choice B is correct. Since $2xy$ is a common factor of each term in the given expression, the expression can be rewritten as $2xy(8x^2y + 7)$.

Choice A is incorrect. This expression is equivalent to $16x^2y^2 + 14xy$.

Choice C is incorrect. This expression is equivalent to $28x^3y^2 + 14xy$.

Choice D is incorrect. This expression is equivalent to $112x^3y^2 + 14xy$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$4a^2 + 20ab + 25b^2$

Which of the following is a factor of the polynomial above?

Answer

- A. $a + b$
- B. $2a + 5b$
- C. $4a + 5b$
- D. $4a + 25b$

Correct Answer: B

Rationale

Choice B is correct. The first and last terms of the polynomial are both squares such that $4a^2 = (2a)^2$ and $25b^2 = (5b)^2$. The second term is twice the product of the square root of the first and last terms: $20ab = 2(2a)(5b)$. Therefore, the polynomial is the square of a binomial such that $4a^2 + 20ab + 25b^2 = (2a + 5b)^2$, and $(2a + 5b)$ is a factor.

Choice A is incorrect and may be the result of incorrectly factoring the polynomial. Choice C is incorrect and may be the result of dividing the second and third terms of the polynomial by their greatest common factor. Choice D is incorrect and may be the result of not factoring the coefficients.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question
If $p = 3x + 4$ and $v = x + 5$, which of the following is equivalent to $pv - 2p + v$?

- Answer**
- A. $3x^2 + 12x + 7$
- B. $3x^2 + 14x + 17$
- C. $3x^2 + 19x + 20$
- D. $3x^2 + 26x + 33$

Correct Answer: B

Rationale
Choice B is correct. It's given that $p = 3x + 4$ and $v = x + 5$. Substituting the values for p and v into the expression $pv - 2p + v$ yields $(3x + 4)(x + 5) - 2(3x + 4) + x + 5$. Multiplying the terms $(3x + 4)(x + 5)$ yields $3x^2 + 4x + 15x + 20$. Using the distributive property to rewrite $-2(3x + 4)$ yields $-6x - 8$. Therefore, the entire expression can be represented as $3x^2 + 4x + 15x + 20 - 6x - 8 + x + 5$. Combining like terms yields $3x^2 + 14x + 17$.

Choice A is incorrect and may result from subtracting, instead of adding, the term $x + 5$. Choice C is incorrect. This is the result of multiplying the terms $(3x + 4)(x + 5)$. Choice D is incorrect and may result from distributing 2, instead of -2 , to the term $3x + 4$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$y = 9\left(\frac{a}{7}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy -plane cross the x -axis, where a , b , and c are positive constants such that $a > 7$ and $b > c$?

Answer

- A. Zero
- B. One
- C. Two
- D. Three

Correct Answer: B

Rationale

Choice B is correct. It’s given that $y = 9\left(\frac{a}{7}\right)^{x+c} - b$, where a , b , and c are positive constants such that $a > 7$ and $b > c$. The graph of any equation in the xy -plane crosses the x -axis when there is a point on the graph for which the value of y is equal to 0, and the value of y changes sign when the graph passes through that point. Since $a > 7$, the base $\frac{a}{7}$ is greater than 1, so the expression $9\left(\frac{a}{7}\right)^{x+c} - b$ increases as x increases. Since b is positive, there is a point on the graph of $y = 9\left(\frac{a}{7}\right)^{x+c} - b$ for which the value of y is equal to 0. When the graph passes through this point, the value of y changes sign from negative to positive. Therefore, the graph of the given equation crosses the x -axis exactly one time.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 7f81d0c3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question
 $x^2 - x - 1 = 0$

What values satisfy the equation above?

Answer
A. $x = 1$ and $x = 2$

B. $x = -\frac{1}{2}$ and $x = \frac{3}{2}$

C. $x = \frac{1 + \sqrt{5}}{2}$ and $x = \frac{1 - \sqrt{5}}{2}$

D. $x = \frac{-1 + \sqrt{5}}{2}$ and $x = \frac{-1 - \sqrt{5}}{2}$

Correct Answer: C

Rationale

Choice C is correct. Using the quadratic formula to solve the given expression yields $x = \frac{-(-1) \pm \sqrt{(-1)^2 - (4)(1)(-1)}}{(2)(1)} = \frac{1 \pm \sqrt{5}}{2}$.

Therefore, $x = \frac{1 + \sqrt{5}}{2}$ and $x = \frac{1 - \sqrt{5}}{2}$ satisfy the given equation.

Choices A and B are incorrect and may result from incorrectly factoring or incorrectly applying the quadratic formula. Choice D is incorrect and may result from a sign error.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$3x^2 - 15x + 18 = 0$

How many distinct real solutions are there to the given equation?

- Answer
- A. Exactly one
 - B. Exactly two
 - C. Infinitely many
 - D. Zero

Correct Answer: B

Rationale

Choice B is correct. The number of solutions to a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $3x^2 - 15x + 18 = 0$, $a = 3$, $b = -15$, and $c = 18$. Substituting 3 for a , -15 for b , and 18 for c in $b^2 - 4ac$ yields $(-15)^2 - 4(3)(18)$, or 9 . Since the value of the discriminant is positive, the given equation has exactly two distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The product of two positive integers is **462**. If the first integer is **5** greater than twice the second integer, what is the smaller of the two integers?

Correct Answer: 14

Rationale

The correct answer is **14**. Let x represent the first integer and y represent the second integer. If the first integer is **5** greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is **462**; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting **462** from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or **-924**, and whose sum is **5**. The two values whose product is **-924** and whose sum is **5** are **33** and **-28**. Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting **33** from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by **2** yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding **14** to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting **14** for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by **14** yields $x = 33$. Therefore, the two integers are **14** and **33**, so the smaller of the two integers is **14**.

Question ID: 128c75e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $g(15a)$ and $g(7a)$?

Correct Answer: 609

Rationale

The correct answer is **609**. It's given that the function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. Substituting $15a$ for x in function g yields $g(15a) = \frac{|15a|}{a} - 14$. This function can be rewritten as $g(15a) = \frac{15|a|}{a} - 14$, or $g(15a) = 15\left(\frac{|a|}{a}\right) - 14$. Since $a < 0$, it follows that $\frac{|a|}{a} = -1$. Substituting -1 for $\frac{|a|}{a}$ in $g(15a) = 15\left(\frac{|a|}{a}\right) - 14$ yields $g(15a) = 15(-1) - 14$, or $g(15a) = -29$. Similarly, substituting $7a$ for x in function g yields $g(7a) = \frac{|7a|}{a} - 14$. This function can be rewritten as $g(7a) = \frac{7|a|}{a} - 14$, or $g(7a) = 7\left(\frac{|a|}{a}\right) - 14$. Since $a < 0$, it again follows that $\frac{|a|}{a} = -1$. Substituting -1 for $\frac{|a|}{a}$ in $g(7a) = 7\left(\frac{|a|}{a}\right) - 14$ yields $g(7a) = 7(-1) - 14$, or $g(7a) = -21$. Therefore, $g(15a) = -29$ and $g(7a) = -21$. Thus, the product of $g(15a)$ and $g(7a)$ is $(-29)(-21)$, or **609**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$y = 7,400(0.87)^x$$

The given equation estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the piece of equipment at the time it was purchased?

Answer

- A. 740
- B. 6,438
- C. 7,400
- D. 8,700

Correct Answer: C

Rationale

Choice C is correct. It’s given that $y = 7,400(0.87)^x$ estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. The estimated value of the piece of equipment at the time it was purchased corresponds to the value of y when $x = 0$. Substituting 0 for x in the equation $y = 7,400(0.87)^x$ yields $y = 7,400(0.87)^0$, which is equivalent to $y = 7,400(1)$, or $y = 7,400$. Therefore, the estimated value, in dollars, of the piece of equipment at the time it was purchased was 7,400.

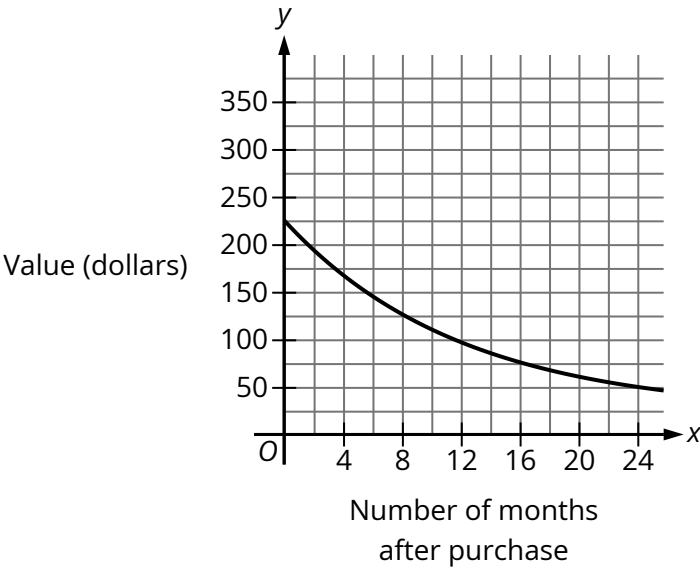
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This was the estimated value, in dollars, of the piece of equipment 1 year after it was purchased.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y -intercept of the graph in this context?

Answer

- A. The estimated value of the tablet was **\$225** when it was purchased.
- B. The estimated value of the tablet **24** months after it was purchased was **\$225**.
- C. The estimated value of the tablet had decreased by **\$225** in the **24** months after it was purchased.
- D. The estimated value of the tablet decreased by approximately **2.25%** each year after it was purchased.

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph shown gives the estimated value y , in dollars, of a tablet as a function of the number of months since it was purchased, x . The y -intercept of a graph is the point at which the graph intersects the y -axis, or when x is 0. The graph shown intersects the y -axis at the point **(0, 225)**. It follows that **0** months after the tablet was purchased, or when the tablet was purchased, the estimated value of the tablet was **225** dollars. Therefore, the best interpretation of the y -intercept is that the estimated value of the tablet was **\$225** when it was purchased.

Choice B is incorrect. The estimated value of the tablet **24** months after it was purchased was **\$50**, not **\$225**.

Choice C is incorrect. The estimated value of the tablet had decreased by **\$225 — \$50**, or **\$175**, not **\$225**, in the **24** months after it was purchased.

Choice D is incorrect and may result from conceptual errors.

Question ID: e312081b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question
 $(x + 5) + (2x - 3)$

Which of the following is equivalent to the given expression?

- Answer
- A. $3x - 2$
- B. $3x + 2$
- C. $3x - 8$
- D. $3x + 8$

Correct Answer: B

Rationale

Choice B is correct. Using the associative and commutative properties of addition, the given expression $(x + 5) + (2x - 3)$ can be rewritten as $(x + 2x) + (5 - 3)$. Adding these like terms results in $3x + 2$.

Choice A is incorrect and may result from adding $(x - 5) + (2x + 3)$. Choice C is incorrect and may result from adding $(x - 5) + (2x - 3)$.

Choice D is incorrect and may result from adding $(x + 5) + (2x + 3)$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

x	$g(x)$
-27	3
-9	0
21	5

The table shows three values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+3}$ and f is a linear function. What is the y-intercept of the graph of $y = f(x)$ in the xy -plane?

Answer

- A. $(0, 36)$
- B. $(0, 12)$
- C. $(0, 4)$
- D. $(0, -9)$

Correct Answer: A

Rationale

Choice A is correct. It's given that the table shows values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+3}$. It's also given that f is a linear function. It follows that an equation that defines f can be written in the form $f(x) = mx + b$, where m represents the slope and b represents the y-coordinate of the y-intercept $(0, b)$ of the graph of $y = f(x)$ in the xy -plane. The slope of the graph of $y = f(x)$ can be found using two points, (x_1, y_1) and (x_2, y_2) , that are on the graph of $y = f(x)$, and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since the table shows values of x and their corresponding values of $g(x)$, substituting values of x and $g(x)$ in the equation $g(x) = \frac{f(x)}{x+3}$ can be used to define function f . Using the first pair of values from the table, $x = -27$ and $g(x) = 3$, yields $3 = \frac{f(-27)}{-27+3}$, or $3 = \frac{f(-27)}{-24}$. Multiplying each side of this equation by -24 yields $-72 = f(-27)$, so the point $(-27, -72)$ is on the graph of $y = f(x)$. Using the second pair of values from the table, $x = -9$ and $g(x) = 0$, yields $0 = \frac{f(-9)}{-9+3}$, or $0 = \frac{f(-9)}{-6}$. Multiplying each side of this equation by -6 yields $0 = f(-9)$, so the point $(-9, 0)$ is on the graph of $y = f(x)$. Substituting $(-27, -72)$ and $(-9, 0)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{0 - (-72)}{-9 - (-27)}$, or $m = 4$. Substituting 4 for m in the equation $f(x) = mx + b$ yields $f(x) = 4x + b$. Since $0 = f(-9)$, substituting -9 for x and 0 for $f(x)$ in the equation $f(x) = 4x + b$ yields $0 = 4(-9) + b$, or $0 = -36 + b$. Adding 36 to both sides of this equation yields $36 = b$. It follows that 36 is the y-coordinate of the y-intercept $(0, b)$ of the graph of $y = f(x)$. Therefore, the y-intercept of the graph of $y = f(x)$ is $(0, 36)$.

Choice B is incorrect. 12 is the y-coordinate of the y-intercept of the graph of $y = g(x)$.

Choice C is incorrect. 4 is the slope of the graph of $y = f(x)$.

Choice D is incorrect. -9 is the x-coordinate of the x-intercept of the graph of $y = f(x)$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which expression is equivalent to $\frac{8x(x-7)-3(x-7)}{2x-14}$, where $x > 7$?

Answer

- A. $\frac{x-7}{5}$
- B. $\frac{8x-3}{2}$
- C. $\frac{8x^2-3x-14}{2x-14}$
- D. $\frac{8x^2-3x-77}{2x-14}$

Correct Answer: B

Rationale

Choice B is correct. The given expression has a common factor of **2** in the denominator, so the expression can be rewritten as $\frac{8x(x-7)-3(x-7)}{2(x-7)}$. The three terms in this expression have a common factor of $(x-7)$. Since it's given that $x > 7$, x can't be equal to **7**, which means $(x-7)$ can't be equal to **0**. Therefore, each term in the expression, $\frac{8x(x-7)-3(x-7)}{2(x-7)}$, can be divided by $(x-7)$, which gives $\frac{8x-3}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$x = 49$

$y = \sqrt{x} + 9$

The graphs of the given equations intersect at the point (x, y) in the xy -plane. What is the value of y ?

Answer

- A. 16
- B. 40
- C. 81
- D. 130

Correct Answer: A

Rationale

Choice A is correct. It's given that the graphs of the given equations intersect at the point (x, y) in the xy -plane. It follows that (x, y) represents a solution to the system consisting of the given equations. The first equation given is $x = 49$. Substituting 49 for x in the second equation given, $y = \sqrt{x} + 9$, yields $y = \sqrt{49} + 9$, which is equivalent to $y = 7 + 9$, or $y = 16$. It follows that the graphs of the given equations intersect at the point $(49, 16)$. Therefore, the value of y is 16.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 91e7ea5e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$h(x) = 2(x - 4)^2 - 32$$

The quadratic function h is defined as shown. In the xy-plane, the graph of $y = h(x)$ intersects the x-axis at the points $(0,0)$ and $(t,0)$, where t is a constant. What is the value of t ?

Answer

- A. 1
- B. 2
- C. 4
- D. 8

Correct Answer: D

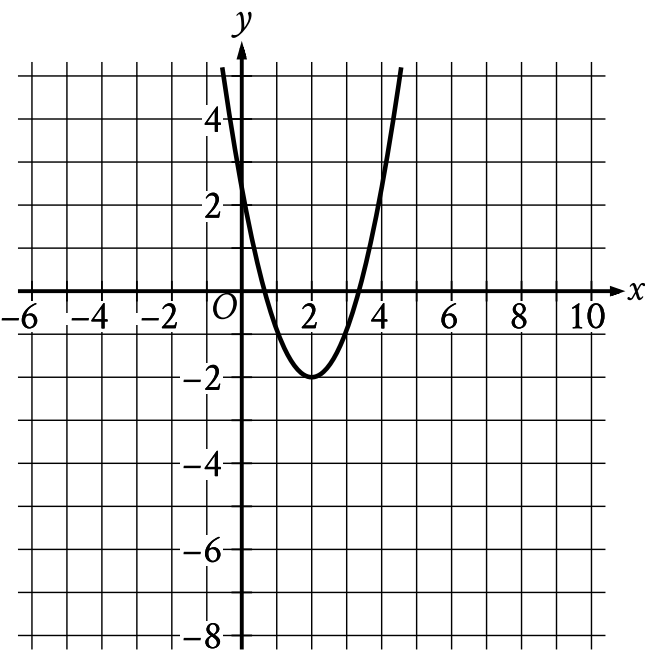
Rationale

Choice D is correct. It’s given that the graph of $y = h(x)$ intersects the x-axis at $(0,0)$ and $(t,0)$, where t is a constant. Since this graph intersects the x-axis when $y = 0$ or when $h(x) = 0$, it follows that $h(0) = 0$ and $h(t) = 0$. If $h(t) = 0$, then $0 = 2(t - 4)^2 - 32$. Adding 32 to both sides of this equation yields $32 = 2(t - 4)^2$. Dividing both sides of this equation by 2 yields $16 = (t - 4)^2$. Taking the square root of both sides of this equation yields $4 = t - 4$. Adding 4 to both sides of this equation yields $8 = t$. Therefore, the value of t is 8.

Choices A, B, and C are incorrect and may result from calculation errors.

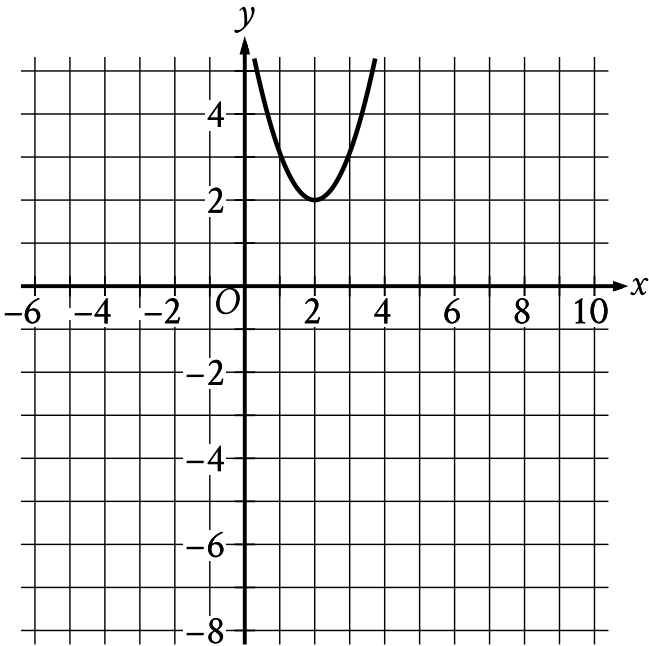
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

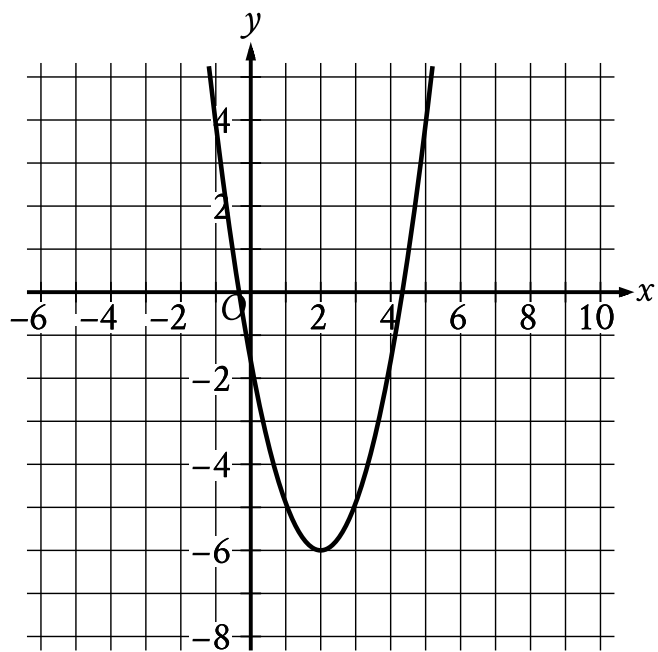


The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

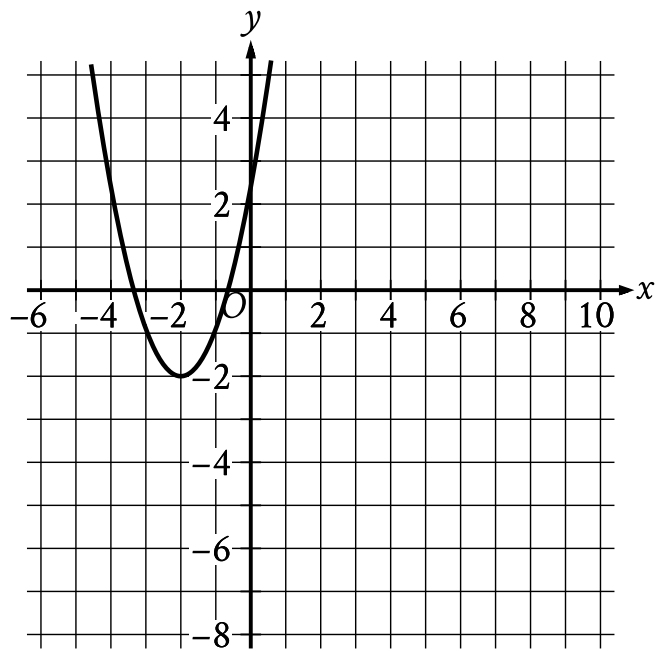
Answer



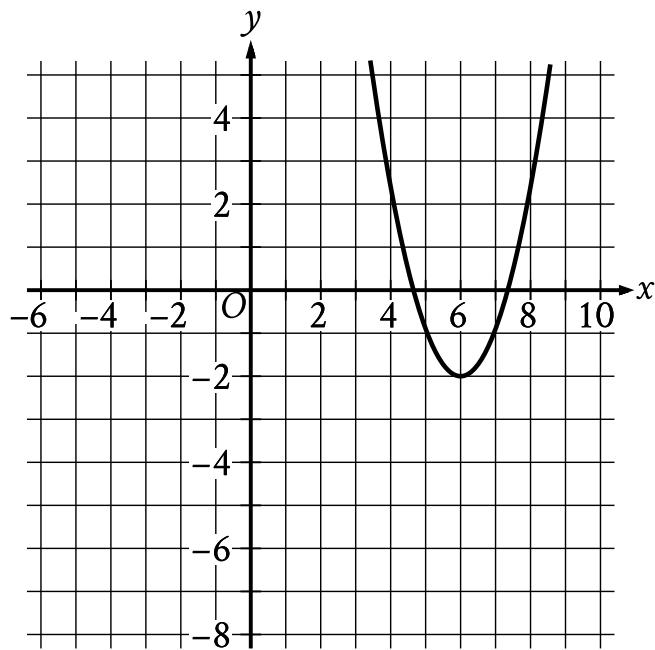
A.



B.



C.



D.

Correct Answer: A

Rationale

Choice A is correct. When a graph is translated up **4** units, each point on the resulting graph is **4** units above the point on the original graph. In other words, the y -value of each point on the graph increases by **4**. The graph shown passes through the points $(1, -1)$, $(2, -2)$, and $(3, -1)$. It follows that when the graph shown is translated up **4** units, the resulting graph will pass through the points $(1, -1 + 4)$, $(2, -2 + 4)$, and $(3, -1 + 4)$. These points are $(1, 3)$, $(2, 2)$, and $(3, 3)$, respectively. Of the given choices, only the graph in choice A passes through the points $(1, 3)$, $(2, 2)$, and $(3, 3)$.

Choice B is incorrect. This is the result of translating the graph down, rather than up, **4** units.

Choice C is incorrect. This is the result of translating the graph left, rather than up, **4** units.

Choice D is incorrect. This is the result of translating the graph right, rather than up, **4** units.

Question ID: 358f18bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = x^2 - 48x + 2,304$

What is the minimum value of the given function?

Correct Answer: 1728

Rationale

The correct answer is **1,728**. The given function can be rewritten in the form $f(x) = a(x - h)^2 + k$, where a is a positive constant and the minimum value, k , of the function occurs when the value of x is h . By completing the square, $f(x) = x^2 - 48x + 2,304$ can be written as $f(x) = x^2 - 48x + \left(\frac{48}{2}\right)^2 + 2,304 - \left(\frac{48}{2}\right)^2$, or $f(x) = (x - 24)^2 + 1,728$. This equation is in the form $f(x) = a(x - h)^2 + k$, where $a = 1$, $h = 24$, and $k = 1,728$. Therefore, the minimum value of the given function is **1,728**.

Question ID: 3a9d60b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$2|4 - x| + 3|4 - x| = 25$

What is the positive solution to the given equation?

Correct Answer: 9

Rationale

The correct answer is 9. The given equation can be rewritten as $5|4 - x| = 25$. Dividing each side of this equation by 5 yields $|4 - x| = 5$. By the definition of absolute value, if $|4 - x| = 5$, then $4 - x = 5$ or $4 - x = -5$. Subtracting 4 from each side of the equation $4 - x = 5$ yields $-x = 1$. Dividing each side of this equation by -1 yields $x = -1$. Similarly, subtracting 4 from each side of the equation $4 - x = -5$ yields $-x = -9$. Dividing each side of this equation by -1 yields $x = 9$. Therefore, since the two solutions to the given equation are -1 and 9, the positive solution to the given equation is 9.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

Answer

- A. $(0, 14)$
- B. $(0, 2)$
- C. $(0, 22)$
- D. $(0, -8)$

Correct Answer: A

Rationale

Choice A is correct. The y -intercept of the graph of $y = f(x)$ in the xy -plane occurs at the point on the graph where $x = 0$. In other words, when $x = 0$, the corresponding value of $f(x)$ is the y -coordinate of the y -intercept. Substituting 0 for x in the given equation yields $f(0) = (-8)(2)^0 + 22$, which is equivalent to $f(0) = (-8)(1) + 22$, or $f(0) = 14$. Thus, when $x = 0$, the corresponding value of $f(x)$ is 14 . Therefore, the y -intercept of the graph of $y = f(x)$ in the xy -plane is $(0, 14)$.

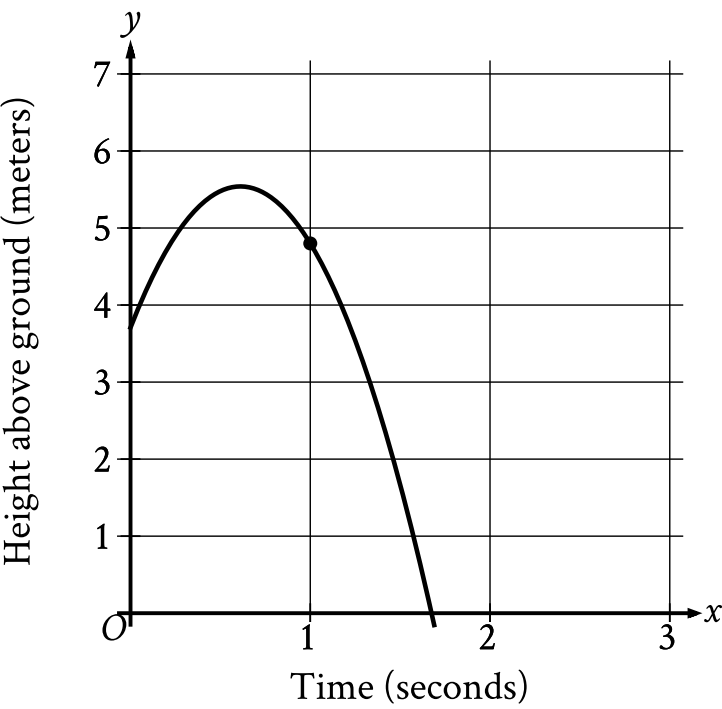
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This could be the y -intercept for $f(x) = (-8)(2)^x$, not $f(x) = (-8)(2)^x + 22$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



The graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point $(1.0, 4.8)$ in this context?

Answer

- A. **1.0** second after being launched, the ball's height above ground is **4.8** meters.
- B. **4.8** seconds after being launched, the ball's height above ground is **1.0** meter.
- C. The ball was launched from an initial height of **1.0** meter with an initial velocity of **4.8** meters per second.
- D. The ball was launched from an initial height of **4.8** meters with an initial velocity of **1.0** meter per second.

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. In the graph shown, the x -axis represents time, in seconds, and the y -axis represents the height of the ball above ground, in meters. It follows that for the marked point $(1.0, 4.8)$, **1.00** represents the time, in seconds, after the ball was launched upward from a platform and **4.80** represents the height of the ball above ground, in meters. Therefore, the best interpretation of the marked point $(1.0, 4.8)$ is **1.00** second after being launched, the ball's height above ground is **4.80** meters.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

An auditorium has seats for 1,800 people. Tickets to attend a show at the auditorium currently cost \$4.00. For each \$1.00 increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

Answer

- A. 4
- B. 7
- C. 14
- D. 18

Correct Answer: B

Rationale

Choice B is correct. It’s given that the situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. Since the coefficient of the x^2 term is negative, the graph of this equation in the xy -plane opens downward and reaches its maximum value at its vertex. If a quadratic equation in the form $y = ax^2 + bx + c$, where a , b , and c are constants, is graphed in the xy -plane, the x -coordinate of the vertex is equal to $-\frac{b}{2a}$. For the equation $y = -100x^2 + 1,400x + 7,200$, $a = -100$, $b = 1,400$, and $c = 7,200$. It follows that the x -coordinate of the vertex is $-\frac{1,400}{2(-100)}$, or 7. Therefore, if the given equation is graphed in the xy -plane, the maximum of the graph occurs at an x -value of 7.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$x^2 - 2x - 9 = 0$$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

Answer

- A. 8
- B. 10
- C. 20
- D. 40

Correct Answer: B

Rationale

Choice B is correct. Adding 9 to each side of the given equation yields $x^2 - 2x = 9$. To complete the square, adding 1 to each side of this equation yields $x^2 - 2x + 1 = 9 + 1$, or $(x - 1)^2 = 10$. Taking the square root of each side of this equation yields $x - 1 = \pm\sqrt{10}$. Adding 1 to each side of this equation yields $x = 1 \pm \sqrt{10}$. Since it's given that one of the solutions to the equation can be written as $1 + \sqrt{k}$, the value of k must be 10.

Alternate approach: It's given that $1 + \sqrt{k}$ is a solution to the given equation. It follows that $x = 1 + \sqrt{k}$. Substituting $1 + \sqrt{k}$ for x in the given equation yields $(1 + \sqrt{k})^2 - 2(1 + \sqrt{k}) - 9 = 0$, or $(1 + \sqrt{k})(1 + \sqrt{k}) - 2(1 + \sqrt{k}) - 9 = 0$. Expanding the products on the left-hand side of this equation yields $1 + 2\sqrt{k} + k - 2 - 2\sqrt{k} - 9 = 0$, or $k - 10 = 0$. Adding 10 to each side of this equation yields $k = 10$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$p(x) + 57 = x^2$$

The given equation relates the value of x and its corresponding value of $p(x)$ for the function p . What is the minimum value of the function p ?

- Answer
- A. ~~−3,249~~
 - B. ~~−57~~
 - C. ~~57~~
 - D. ~~3,249~~

Correct Answer: B

Rationale

Choice B is correct. For a quadratic function defined by an equation of the form $p(x) = a(x - h)^2 + k$, where a , h , and k are constants and $a > 0$, the minimum value of the function is k . Subtracting 57 from both sides of the given equation yields $p(x) = x^2 - 57$. This function is in the form $p(x) = a(x - h)^2 + k$, where $a = 1$, $h = 0$, and $k = -57$. Therefore, the minimum value of the function p is -57 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

In the xy -plane, a line with equation $2y = 4.5$ intersects a parabola at exactly one point. If the parabola has equation $y = -4x^2 + bx$, where b is a positive constant, what is the value of b ?

Correct Answer: 6

Rationale

The correct answer is **6**. It's given that a line with equation $2y = 4.5$ intersects a parabola with equation $y = -4x^2 + bx$, where b is a positive constant, at exactly one point in the xy -plane. It follows that the system of equations consisting of $2y = 4.5$ and $y = -4x^2 + bx$ has exactly one solution. Dividing both sides of the equation of the line by **2** yields $y = 2.25$. Substituting **2.25** for y in the equation of the parabola yields $2.25 = -4x^2 + bx$. Adding $4x^2$ and subtracting bx from both sides of this equation yields $4x^2 - bx + 2.25 = 0$. A quadratic equation in the form of $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one solution when the discriminant, $b^2 - 4ac$, is equal to zero. Substituting **4** for a and **2.25** for c in the expression $b^2 - 4ac$ and setting this expression equal to **0** yields $b^2 - 4(4)(2.25) = 0$, or $b^2 - 36 = 0$. Adding **36** to each side of this equation yields $b^2 = 36$. Taking the square root of each side of this equation yields $b = \pm 6$. It's given that b is positive, so the value of b is **6**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$f(x) = 9,000(0.66)^x$$

The given function f models the number of advertisements a company sent to its clients each year, where x represents the number of years since **1997**, and $0 \leq x \leq 5$. If $y = f(x)$ is graphed in the xy -plane, which of the following is the best interpretation of the y -intercept of the graph in this context?

Answer

- A. The minimum estimated number of advertisements the company sent to its clients during the **5** years was **1,708**.
- B. The minimum estimated number of advertisements the company sent to its clients during the **5** years was **9,000**.
- C. The estimated number of advertisements the company sent to its clients in **1997** was **1,708**.
- D. The estimated number of advertisements the company sent to its clients in **1997** was **9,000**.

Correct Answer: D

Rationale

Choice D is correct. The y -intercept of a graph in the xy -plane is the point where $x = 0$. For the given function f , the y -intercept of the graph of $y = f(x)$ in the xy -plane can be found by substituting **0** for x in the equation $y = 9,000(0.66)^x$, which gives $y = 9,000(0.66)^0$. This is equivalent to $y = 9,000(1)$, or $y = 9,000$. Therefore, the y -intercept of the graph of $y = f(x)$ is **(0, 9,000)**. It's given that the function f models the number of advertisements a company sent to its clients each year. Therefore, $f(x)$ represents the estimated number of advertisements the company sent to its clients each year. It's also given that x represents the number of years since **1997**. Therefore, $x = 0$ represents **0** years since **1997**, or **1997**. Thus, the best interpretation of the y -intercept of the graph of $y = f(x)$ is that the estimated number of advertisements the company sent to its clients in **1997** was **9,000**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $(9x^3 + 5x + 7) + (6x^3 + 5x^2 - 5)$?

Answer

- A. $15x^6 + 5x^2 - 5x - 35$
- B. $15x^3 + 10x^2 + 2$
- C. $15x^6 + 5x^2 + 5x + 2$
- D. $15x^3 + 5x^2 + 5x + 2$

Correct Answer: D

Rationale

Choice D is correct. The given expression can be rewritten as $(9x^3 + 6x^3) + 5x^2 + 5x + (7 - 5)$. Combining like terms in this expression yields $15x^3 + 5x^2 + 5x + 2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 2c6f214f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The first term of a sequence is **9**. Each term after the first is **4** times the preceding term. If w represents the n th term of the sequence, which equation gives w in terms of n ?

Answer

- A. $w = 4(9^n)$
- B. $w = 4(9^{n-1})$
- C. $w = 9(4^n)$
- D. $w = 9(4^{n-1})$

Correct Answer: D

Rationale

Choice D is correct. Since w represents the n th term of the sequence and **9** is the first term of the sequence, the value of w is **9** when the value of n is **1**. Since each term after the first is **4** times the preceding term, the value of w is **9(4)** when the value of n is **2**. Therefore, the value of w is **9(4)(4)**, or **9(4)²**, when the value of n is **3**. More generally, the value of w is **9(4ⁿ⁻¹)** for a given value of n . Therefore, the equation $w = 9(4^{n-1})$ gives w in terms of n .

Choice A is incorrect. This equation describes a sequence for which the first term is **36**, rather than **9**, and each term after the first is **9**, rather than **4**, times the preceding term.

Choice B is incorrect. This equation describes a sequence for which the first term is **4**, rather than **9**, and each term after the first is **9**, rather than **4**, times the preceding term.

Choice C is incorrect. This equation describes a sequence for which the first term is **36**, rather than **9**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = a(2.2^x + 2.2^b)$, where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f , where k and m are constants. Which of the following equations displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient?

- I. $g(x) = a(2.2^x + k)$
- II. $h(x) = a(2.2)^x + m$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: D

Rationale

Choice D is correct. A y -intercept of a graph in the xy -plane is a point where the graph intersects the y -axis, or a point where $x = 0$. Substituting 0 for x in the equation defining function f yields $f(0) = a(2.2^0 + 2.2^b)$, or $f(0) = a(1 + 2.2^b)$. So, the y -coordinate of the y -intercept of the graph is $a(1 + 2.2^b)$, or equivalently, $a + a(2.2)^b$. It's given that function g is equivalent to function f , where $0 < a < b$. It follows that $k = 2.2^b$. Since $a(2.2)^b$ can't be equal to 0 , the coefficient a can't be equal to $a + a(2.2)^b$. Since $0 < a$, the constant k , which is equal to 2.2^b , can't be equal to $a + a(2.2)^b$. Therefore, function g doesn't display the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient. It's also given that function h is equivalent to function f , where $0 < a < b$. The equation defining f can be rewritten as $f(x) = a(2.2)^x + a(2.2)^b$. It follows that $m = a(2.2)^b$. Since $a(2.2)^b$ can't be equal to 0 , the coefficient a can't be equal to $a + a(2.2)^b$. Since $0 < a$, the constant m , which is equal to $a(2.2)^b$, can't be equal to $a + a(2.2)^b$. Therefore, function h doesn't display the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient. Thus, neither function g nor function h displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

A physics class is planning an experiment about a toy rocket. The equation $y = -16(x - 5.6)^2 + 502$ gives the estimated height y , in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy -plane?

Answer

- A. This toy rocket reaches an estimated maximum height of **502** feet **16** seconds after it is launched into the air.
- B. This toy rocket reaches an estimated maximum height of **502** feet **5.6** seconds after it is launched into the air.
- C. This toy rocket reaches an estimated maximum height of **16** feet **502** seconds after it is launched into the air.
- D. This toy rocket reaches an estimated maximum height of **5.6** feet **502** seconds after it is launched into the air.

Correct Answer: B

Rationale

Choice B is correct. The vertex of the graph of a quadratic equation is where it reaches its minimum or maximum value. When a quadratic equation is written in the form $y = a(x - h)^2 + k$, the vertex of the parabola represented by the equation is $(x, y) = (h, k)$. In the given equation $y = -16(x - 5.6)^2 + 502$, the value of h is **5.6** and the value of k is **502**. It follows that the vertex of the graph of this equation in the xy -plane is $(x, y) = (5.6, 502)$. Additionally, since $a = -16$ in the given equation, the graph of the quadratic equation opens down, and the vertex represents a maximum. It's given that the value of y represents the estimated height, in feet, of the toy rocket x seconds after it is launched into the air. Therefore, this toy rocket reaches an estimated maximum height of **502** feet **5.6** seconds after it is launched into the air.

Choice A is incorrect. The **16** in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice C is incorrect. The **16** in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice D is incorrect. This is an interpretation of the vertex of the graph of the equation $y = -16(x - 502)^2 + 5.6$, not of the equation $y = -16(x - 5.6)^2 + 502$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$x - y = 1$$
$$x + y = x^2 - 3$$

Which ordered pair is a solution to the system of equations above?

- Answer
- A. $(1 + \sqrt{3}, \sqrt{3})$
- B. $(\sqrt{3}, -\sqrt{3})$
- C. $(1 + \sqrt{5}, \sqrt{5})$
- D. $(\sqrt{5}, -1 + \sqrt{5})$

Correct Answer: A

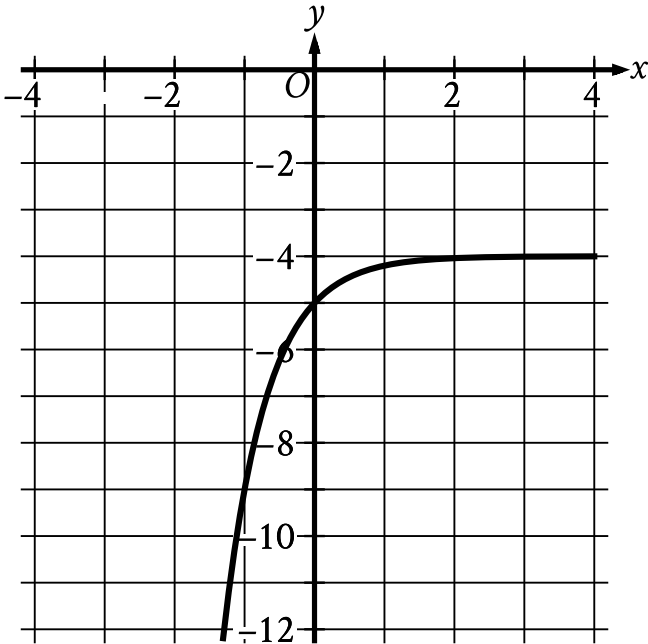
Rationale

Choice A is correct. The solution to the given system of equations can be found by solving the first equation for x , which gives $x = y + 1$, and substituting that value of x into the second equation which gives $y + 1 + y = (y + 1)^2 - 3$. Rewriting this equation by adding like terms and expanding $(y + 1)^2$ gives $2y + 1 = y^2 + 2y - 2$. Subtracting $2y$ from both sides of this equation gives $1 = y^2 - 2$. Adding 2 to both sides of this equation gives $3 = y^2$. Therefore, it follows that $y = \pm\sqrt{3}$. Substituting $\sqrt{3}$ for y in the first equation yields $x - \sqrt{3} = 1$. Adding $\sqrt{3}$ to both sides of this equation yields $x = 1 + \sqrt{3}$. Therefore, the ordered pair $(1 + \sqrt{3}, \sqrt{3})$ is a solution to the given system of equations.

Choice B is incorrect. Substituting $\sqrt{3}$ for x and $-\sqrt{3}$ for y in the first equation yields $\sqrt{3} - (-\sqrt{3}) = 1$, or $2\sqrt{3} = 1$, which isn't a true statement. Choice C is incorrect. Substituting $1 + \sqrt{5}$ for x and $\sqrt{5}$ for y in the second equation yields $(1 + \sqrt{5}) + \sqrt{5} = (1 + \sqrt{5})^2 - 3$, or $1 + 2\sqrt{5} = 2\sqrt{5} + 3$, which isn't a true statement. Choice D is incorrect. Substituting $\sqrt{5}$ for x and $(-1 + \sqrt{5})$ for y in the second equation yields $\sqrt{5} + (-1 + \sqrt{5}) = (\sqrt{5})^2 - 3$, or $2\sqrt{5} - 1 = 2$, which isn't a true statement.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



What is the y-intercept of the graph shown?

- Answer**
- A. $(-1, -9)$
 - B. $(0, -5)$
 - C. $(0, -4)$
 - D. $(0, 0)$

Correct Answer: B

Rationale

Choice B is correct. The y-intercept of a graph in the xy -plane is the point (x, y) on the graph where $x = 0$. At $x = 0$, the corresponding value of y is -5 . Therefore, the y-intercept of the graph shown is $(0, -5)$.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the y-intercept of a graph in the xy -plane that intersects the y -axis at $y = -4$, not $y = -5$.

Choice D is incorrect. This is the y-intercept of a graph in the xy -plane that intersects the y -axis at $y = 0$, not $y = -5$.

Question ID: ad2ec615

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which of the following is equivalent to the expression $x^4 - x^2 - 6$?

Answer

- A. $(x^2 + 1)(x^2 - 6)$
- B. $(x^2 + 2)(x^2 - 3)$
- C. $(x^2 + 3)(x^2 - 2)$
- D. $(x^2 + 6)(x^2 - 1)$

Correct Answer: B

Rationale

Choice B is correct. The term x^4 can be factored as $(x^2)(x^2)$. Factoring -6 as $(2)(-3)$ yields values that add to -1 , the coefficient of x^2 in the expression.

Choices A, C, and D are incorrect and may result from finding factors of -6 that don't add to the coefficient of x^2 in the original expression.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question
 $(2x + 5)^2 - (x - 2) + 2(x + 3)$

Which of the following is equivalent to the expression above?

- Answer**
- A. $4x^2 + 21x + 33$
 - B. $4x^2 + 21x + 29$
 - C. $4x^2 + x + 29$
 - D. $4x^2 + x + 33$

Correct Answer: A

Rationale

Choice A is correct. The given expression can be rewritten as $(2x + 5)^2 + (-1)(x - 2) + 2(x + 3)$. Applying the distributive property, the expression $(-1)(x - 2) + 2(x + 3)$ can be rewritten as $-1(x) + (-1)(-2) + 2(x) + 2(3)$, or $-x + 2 + 2x + 6$. Adding like terms yields $x + 8$. Substituting $x + 8$ for $(-1)(x - 2) + 2(x + 3)$ in the given expression yields $(2x + 5)^2 + x + 8$. By the rules of exponents, the expression $(2x + 5)^2$ is equivalent to $(2x + 5)(2x + 5)$. Applying the distributive property, this expression can be rewritten as $2x(2x) + 2x(5) + 5(2x) + 5(5)$, or $4x^2 + 10x + 10x + 25$. Adding like terms gives $4x^2 + 20x + 25$. Substituting $4x^2 + 20x + 25$ for $(2x + 5)^2$ in the rewritten expression yields $4x^2 + 20x + 25 + x + 8$, and adding like terms yields $4x^2 + 21x + 33$.

Choices B, C, and D are incorrect. Choices C and D may result from rewriting the expression $(2x + 5)^2$ as $4x^2 + 25$, instead of as $4x^2 + 20x + 25$. Choices B and C may result from rewriting the expression $-(x - 2)$ as $-x - 2$, instead of $-x + 2$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

Time (years)	Total amount (dollars)
0	604.00
1	606.42
2	608.84

Rosa opened a savings account at a bank. The table shows the exponential relationship between the time t , in years, since Rosa opened the account and the total amount n , in dollars, in the account. If Rosa made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n ?

Answer

- A. $n = (1 + 604)^t$
- B. $n = (1 + 0.004)^t$
- C. $n = 604(1 + 0.004)^t$
- D. $n = 0.004(1 + 604)^t$

Correct Answer: C

Rationale

Choice C is correct. It’s given that the relationship between t and n is exponential. The table shows that the value of n increases as the value of t increases. Therefore, the relationship between t and n can be represented by an increasing exponential equation of the form $n = a(1 + b)^t$, where a and b are positive constants. The table shows that when $t = 0, n = 604$. Substituting 0 for t and 604 for n in the equation $n = a(1 + b)^t$ yields $604 = a(1 + b)^0$, which is equivalent to $604 = a(1)$, or $604 = a$. Substituting 604 for a in the equation $n = a(1 + b)^t$ yields $n = 604(1 + b)^t$. The table also shows that when $t = 1, n = 606.42$. Substituting 1 for t and 606.42 for n in the equation $n = 604(1 + b)^t$ yields $606.42 = 604(1 + b)^1$, or $606.42 = 604(1 + b)$. Dividing both sides of this equation by 604 yields approximately $1.004 = 1 + b$. Subtracting 1 from both sides of this equation yields that the value of b is approximately 0.004. Substituting 0.004 for b in the equation $n = 604(1 + b)^t$ yields $n = 604(1 + 0.004)^t$. Therefore, of the choices, choice C best represents the relationship between t and n .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$

The equation above is true for all x, where a and b are constants. What is the value of ab ?

Answer

- A. 18
- B. 20
- C. 24
- D. 40

Correct Answer: C

Rationale

Choice C is correct. If the equation is true for all x, then the expressions on both sides of the equation will be equivalent. Multiplying the polynomials on the left-hand side of the equation gives $5ax^3 - abx^2 + 4ax + 15x^2 - 3bx + 12$. On the right-hand side of the equation, the only x^2 -term is $-9x^2$. Since the expressions on both sides of the equation are equivalent, it follows that $-abx^2 + 15x^2 = -9x^2$, which can be rewritten as $(-ab + 15)x^2 = -9x^2$. Therefore, $-ab + 15 = -9$, which gives $ab = 24$.

Choice A is incorrect. If $ab = 18$, then the coefficient of x^2 on the left-hand side of the equation would be $-18 + 15 = -3$, which doesn't equal the coefficient of x^2 , -9 , on the right-hand side. Choice B is incorrect. If $ab = 20$, then the coefficient of x^2 on the left-hand side of the equation would be $-20 + 15 = -5$, which doesn't equal the coefficient of x^2 , -9 , on the right-hand side. Choice D is incorrect. If $ab = 40$, then the coefficient of x^2 on the left-hand side of the equation would be $-40 + 15 = -25$, which doesn't equal the coefficient of x^2 , -9 , on the right-hand side.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$x^2 - 12x + 27 = 0$$

How many distinct real solutions does the given equation have?

Answer

- A. Exactly two
- B. Exactly one
- C. Zero
- D. Infinitely many

Correct Answer: A

Rationale

Choice A is correct. The number of solutions of a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $x^2 - 12x + 27 = 0$, $a = 1$, $b = -12$, and $c = 27$. Substituting these values for a , b , and c in $b^2 - 4ac$ yields $(-12)^2 - 4(1)(27)$, or 36 . Since the value of its discriminant is positive, the given equation has exactly two distinct real solutions.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

x	$g(x)$
-1	25
0	1
1	$\frac{1}{25}$
2	$\frac{1}{625}$

For the exponential function g , the table shows four values of x and their corresponding values of $g(x)$. Which equation defines g ?

Answer

- A. $g(x) = -25^x$
- B. $g(x) = -\left(\frac{1}{25}\right)^x$
- C. $g(x) = 25^x$
- D. $g(x) = \left(\frac{1}{25}\right)^x$

Correct Answer: D

Rationale

Choice D is correct. It's given that function g is exponential. Therefore, an equation defining g can be written in the form $g(x) = a(b)^x$, where a and b are constants. The table shows that when $x = 0$, $g(x) = 1$. Substituting 0 for x and 1 for $g(x)$ in the equation $g(x) = a(b)^x$ yields $1 = a(b)^0$, which is equivalent to $1 = a$. Substituting 1 for a in the equation $g(x) = a(b)^x$ yields $g(x) = (b)^x$. The table also shows that when $x = 1$, $g(x) = \frac{1}{25}$. Substituting 1 for x and $\frac{1}{25}$ for $g(x)$ in the equation $g(x) = (b)^x$ yields $\frac{1}{25} = (b)^1$, which is equivalent to $\frac{1}{25} = b$. Substituting $\frac{1}{25}$ for b in the equation $g(x) = (b)^x$ yields $g(x) = \left(\frac{1}{25}\right)^x$.

Choice A is incorrect. For this function, $g(1)$ is equal to -25 , not $\frac{1}{25}$.

Choice B is incorrect. For this function, $g(1)$ is equal to $-\frac{1}{25}$, not $\frac{1}{25}$.

Choice C is incorrect. For this function, $g(1)$ is equal to 25 , not $\frac{1}{25}$.

Question ID: a05bd3a4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which of the following expressions is equivalent to $x^2 - 5$?

Answer

- A. $(x + \sqrt{5})^2$
- B. $(x - \sqrt{5})^2$
- C. $(x + \sqrt{5})(x - \sqrt{5})$
- D. $(x + 5)(x - 1)$

Correct Answer: C

Rationale

Choice C is correct. The expression can be written as a difference of squares $x^2 - y^2$, which can be factored as $(x + y)(x - y)$. Here, $y^2 = 5$, so $y = \sqrt{5}$, and the expression therefore factors as $(x + \sqrt{5})(x - \sqrt{5})$.

Choices A and B are incorrect and may result from misunderstanding how to factor a difference of squares. Choice D is incorrect; $(x + 5)(x - 1)$ can be rewritten as $x^2 + 4x - 5$, which is not equivalent to the original expression.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which of the following expressions is(are) a factor of $3x^2 + 20x - 63$?

- I. $x - 9$
- II. $3x - 7$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: B

Rationale

Choice B is correct. The given expression can be factored by first finding two values whose sum is **20** and whose product is **3(−63)**, or **−189**. Those two values are **27** and **−7**. It follows that the given expression can be rewritten as $3x^2 + 27x - 7x - 63$. Since the first two terms of this expression have a common factor of **3x** and the last two terms of this expression have a common factor of **−7**, this expression can be rewritten as $3x(x + 9) - 7(x + 9)$. Since the two terms of this expression have a common factor of **(x + 9)**, it can be rewritten as $(3x - 7)(x + 9)$. Therefore, expression II, **3x − 7**, is a factor of $3x^2 + 20x - 63$, but expression I, **x − 9**, is not a factor of $3x^2 + 20x - 63$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

If $\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$ for all positive values of x, what is the value of $\frac{a}{b}$?

Rationale

The correct answer is $\frac{7}{6}$. The value of $\frac{a}{b}$ can be found by first rewriting the left-hand side of the given equation as $\frac{x^{\frac{5}{2}}}{x^{\frac{4}{3}}}$. Using the properties of exponents, this expression can be rewritten as $x^{\left(\frac{5}{2} - \frac{4}{3}\right)}$. This expression can be rewritten by subtracting the fractions in the exponent, which yields $x^{\frac{7}{6}}$. Thus, $\frac{a}{b}$ is $\frac{7}{6}$. Note that 7/6, 1.166, and 1.167 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

x	$h(x)$
0	1.23
2	1.54
4	1.94

The table shows the exponential relationship between the number of years, x , since Hana started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

Answer

- A. $h(x) = 1.12(0.23)^x$
- B. $h(x) = 1.12(1.23)^x$
- C. $h(x) = 1.23(0.12)^x$
- D. $h(x) = 1.23(1.12)^x$

Correct Answer: D

Rationale

Choice D is correct. The table shows an increasing exponential relationship between the number of years, x , since Hana started training in pole vault and the estimated height $h(x)$, in meters, of her best pole vault for that year. The relationship can be written as $h(x) = Ca^x$, where C and a are positive constants. It's given that when $x = 0$, $h(x) = 1.23$. Substituting 0 for x and 1.23 for $h(x)$ in $h(x) = Ca^x$ yields $1.23 = Ca^0$, or $1.23 = C$. Substituting 1.23 for C in $h(x) = Ca^x$ yields $h(x) = 1.23a^x$. It's also given that when $x = 2$, $h(x) = 1.54$. Substituting 2 for x and 1.54 for $h(x)$ in $h(x) = 1.23a^x$ yields $1.54 = 1.23a^2$. Dividing each side of this equation by 1.23 yields $\frac{1.54}{1.23} = \frac{1.23a^2}{1.23}$, or a^2 is approximately equal to 1.252. Since a is positive, a is approximately equal to $\sqrt{1.252}$, or 1.12. Substituting 1.12 for a in $h(x) = 1.23a^x$ yields $h(x) = 1.23(1.12)^x$.

Choice A is incorrect. When $x = 0$, the value of $h(x)$ in this function is equal to 1.12 rather than 1.23, and it is decreasing rather than increasing.

Choice B is incorrect. When $x = 0$, the value of $h(x)$ in this function is equal to 1.12 rather than 1.23.

Choice C is incorrect. This function is decreasing rather than increasing.

Question ID: b8f13a3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 12$ has a y -intercept at $(0, -\frac{75}{7})$. The product of a and b is $\frac{320}{7}$. What is the value of a ?

Correct Answer: 20

Rationale

The correct answer is **20**. It's given that $f(x) = -a^x + b$. Substituting $-a^x + b$ for $f(x)$ in the equation $y = f(x) - 12$ yields $y = -a^x + b - 12$. It's given that the y -intercept of the graph of $y = f(x) - 12$ is $(0, -\frac{75}{7})$. Substituting 0 for x and $-\frac{75}{7}$ for y in the equation $y = -a^x + b - 12$ yields $-\frac{75}{7} = -a^0 + b - 12$, which is equivalent to $-\frac{75}{7} = -1 + b - 12$, or $-\frac{75}{7} = b - 13$. Adding 13 to both sides of this equation yields $\frac{16}{7} = b$. It's given that the product of a and b is $\frac{320}{7}$, or $ab = \frac{320}{7}$. Substituting $\frac{16}{7}$ for b in this equation yields $(a)(\frac{16}{7}) = \frac{320}{7}$. Dividing both sides of this equation by $\frac{16}{7}$ yields $a = 20$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = (x - 44)(x - 46)$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

Answer

- A. 46
- B. 45
- C. 44
- D. -1

Correct Answer: B

Rationale

Choice B is correct. It's given that $f(x) = (x - 44)(x - 46)$, which can be rewritten as $f(x) = x^2 - 90x + 2,024$. Since the coefficient of the x^2 -term is positive, the graph of $y = f(x)$ in the xy -plane opens upward and reaches its minimum value at its vertex. For an equation in the form $f(x) = ax^2 + bx + c$, where a , b , and c are constants, the x -coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x) = x^2 - 90x + 2,024$, $a = 1$, $b = -90$, and $c = 2,024$. It follows that the x -coordinate of the vertex is $-\frac{(-90)}{2(1)}$, or 45. Therefore, $f(x)$ reaches its minimum when the value of x is 45.

Choice A is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = f(x)$ in the xy -plane.

Choice C is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = f(x)$ in the xy -plane.

Choice D is incorrect. This is the y -coordinate of the vertex of the graph of $y = f(x)$ in the xy -plane.